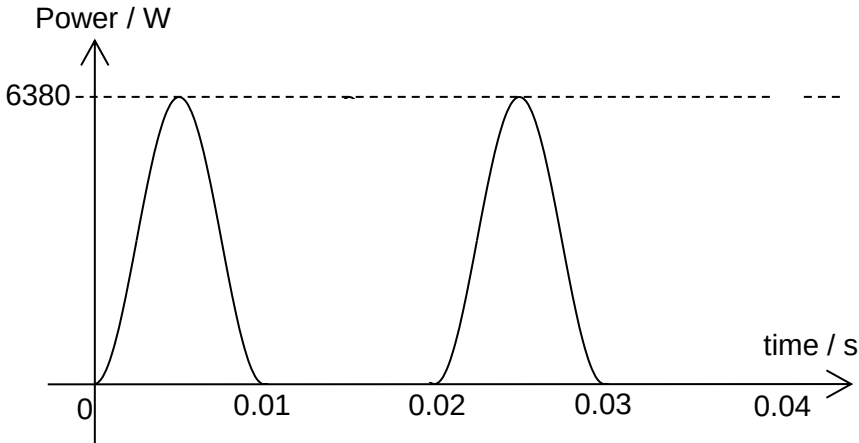


Qn		Answer	Mark
5	(a)(i)	Resistance of copper, R_{copper} $= \frac{(1.60 \times 10^{-8})(3.0)}{\pi (0.60 \times 10^{-3} + 1.78 \times 10^{-5})^2 - \pi (0.60 \times 10^{-3})^2}$ $= 0.705 \Omega$ Effective resistance $= \left(\frac{1}{0.236} + \frac{1}{0.705} \right)^{-1}$ $= 0.176 \Omega \text{ (show intermediate value of higher sf)}$ $= 0.18 \Omega$	M1 A1
	(a)(ii)	As current increases, power dissipated increases. Heat is generated and hence equilibrium temperature increases. Hence, the lattice ions vibrate more vigorously, hindering the flow of 'charge carriers', therefore increasing the resistance.	B1 B1

	(b)(i) Using $V_s/V_p = N_s/N_p$, $(\max V_s)/240\sqrt{2} = 1/10$ $\max V_s = 33.9 \text{ V}$	M1 A1
	(b)(ii) period = 0.02 s maximum power $P_0 = \frac{V_0^2}{R} = \frac{(33.9)^2}{0.18} = 6380 \text{ W}$  1 mark for correct shape 1 mark for correct labelling peak power and period	B2

	(b)(iii)	$\langle V^2 \rangle = \frac{1}{2} \frac{V_0^2 (T/2)}{T} = \frac{1}{4} V_0^2 \Rightarrow V_{rms} = \sqrt{\langle V^2 \rangle} = \frac{1}{2} V_0$ $\langle P \rangle = V_{rms}^2 / R = V_0^2 / 4R$ $\langle P \rangle = (33.9)^2 / 4(0.18) = 1600 \text{ W}$ <u>OR</u> $P_0 = \frac{V_0^2}{R} = \frac{(33.9)^2}{0.18} = 6380 \text{ W}$ Mean power for sinusoidal graph = $\frac{1}{2} P_0$ Since it is half-wave rectified, mean power for half-wave rectified graph = $\frac{1}{4} P_0 = \frac{1}{4} (6380) = 1600 \text{ W}$ If students use graphical method, deduct 1 mark if students <u>did not provide clear steps</u> on how V_{rms} and mean P are determined. There should be clear steps leading to the final answer.	C1 M1 A1 C1 M1 A1
	(c)	A voltage of 24 V steady direct current voltage has the same value as the rms secondary voltage of 24 V of the alternating current. <u>The rms voltage of the half-wave rectified voltage is lower (16.9 V).</u> Hence, the power increases when 24 V direct current voltage is used.	B1

Questions	Answers	Marks
6 (c)	Electric field strength at a point is defined by the force per unit	B1